
Computing GDP

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This note explains the difference between nominal and real GDP, why there exist different types of real GDP measures, and how they are computed. In section 1, we set up a simple economy and explain how nominal GDP is computed. To keep track of how a country's production changes over time, economists have invented real GDP measures. The simplest one is real GDP at constant prices of a certain base year; a more useful real GDP measure is chain-weighted real GDP. These GDP measures are explained in section 2. To compare a country's production with the production of another country, economists use GDP at Purchasing Power Parity (PPP), which is explained in section 3.

1 Nominal GDP

Consider a country that produces two final goods in three consecutive years. Denote the quantity of good i produced in year t by $q_{i,t}$, and the price of good i in year t by $p_{i,t}$. The quantities q and the prices p of the final goods in each of the three years are given in Table 1.

Table 1: Quantities and prices

	Year 1	Year 2	Year 3
Quantity			
• of good 1	100	80	90
• of good 2	100	110	110
Price			
• of good 1	4	6	12
• of good 2	10	10	10

Nominal GDP in year t , $P_t Y_t$, is the value of all the final goods produced in year t , valued at their prices in year t :

$$P_t Y_t = \sum_i p_{i,t} q_{i,t} = p_{1,t} q_{1,t} + p_{2,t} q_{2,t}$$

For instance, nominal GDP in year 1 is equal to

$$P_1 Y_1 = p_{1,1} q_{1,1} + p_{2,1} q_{2,1} = 4 \cdot 100 + 10 \cdot 100 = 1400$$

In the same way, we find that $P_2 Y_2 = 1580$ and $P_3 Y_3 = 2180$.

The growth rate of nominal GDP from year t to year $t + 1$, g_{t+1} , is equal to

$$g_{t+1} = \frac{P_{t+1}Y_{t+1} - P_tY_t}{P_tY_t}$$

For instance, the growth rate of nominal GDP from year 1 to year 2 is equal to

$$g_2 = \frac{P_2Y_2 - P_1Y_1}{P_1Y_1} = \frac{1580 - 1400}{1400} \approx 0.129 = 12.9\%$$

In the same way, we find that the growth rate of nominal GDP from year 2 to year 3 is (approximately) 38.0%. Nominal GDP and the growth rate of nominal GDP are summarized in Table 2.

Table 2: Nominal GDP

	Year 1	Year 2	Year 3
Nominal GDP			
• level	1400	1580	2180
• growth rate	-	12.9%	38.0%

Nominal GDP in year t depends on the prices and the produced quantities in year t . Often, however, we want to keep track of how produced quantities change over time. The growth rate of nominal GDP is not a good measure for this, as nominal GDP does not only change when produced quantities change but also when prices change. That is why economists have invented real GDP measures, which is the topic of the next section.

2 Real GDP

Real GDP in year t , Y_t , is a measure of the produced quantities in year t , such that the *growth rate* of real GDP is not affected by changes in the prices of the different goods and is therefore a good measure of the growth in produced quantities. We first look at real GDP at constant prices of a certain base year; we then look at chain-weighted real GDP, which is more complicated to compute, but more useful and therefore in many countries the most common way to measure real GDP.

2.1 Real GDP at constant prices of a certain base year

Suppose that we want to find out whether the amount of final goods, taking all final goods together, has increased or decreased between year 1 and year 2.

One way how we could do this is by simply adding up all the quantities produced in year 1, $\sum_i q_{i,1} = q_{1,1} + q_{2,1}$, and comparing this with the sum of the quantities in year 2, $\sum_i q_{i,2} = q_{1,2} + q_{2,2}$. But then good 1 would carry the same weight in our computation as good 2, even though in both years good 1 was substantially cheaper than good 2. That does not seem reasonable.

So a better way is to use a set of prices to value each of the final goods, where cheap goods such as good 1 get a lower price and therefore less weight in our computations, and more expensive goods such as good 2 get a higher price and carry more weight. However, we then have to make sure that we use the same set of prices in year 1 and year 2. Otherwise, our results may be driven by the fact that we use different prices in the two years, rather than by the different quantities that are produced - which is the very reason why we cannot use nominal GDP to compare the amount of produced quantities in two different years.

There are many price sets which we could use for this purpose. But a logical choice is to use the prices of a specific year. This year is called the base year.

In principle, we can choose any year as a base year. Let us take year 1 as base year, for instance. The prices which we use in all three years to compute real GDP are then 4 for good 1 and 10 for good 2. Real GDP in the three years is then equal to

$$\begin{aligned} Y_1^1 &= \sum_i p_{i,1} q_{i,1} = 4 \cdot 100 + 10 \cdot 100 = 1400 \\ Y_2^1 &= \sum_i p_{i,1} q_{i,2} = 4 \cdot 80 + 10 \cdot 110 = 1420 \\ Y_3^1 &= \sum_i p_{i,1} q_{i,3} = 4 \cdot 90 + 10 \cdot 110 = 1460 \end{aligned}$$

where the superscript denotes the base year. Note that real GDP in year 1 is the same as nominal GDP in year 1. Of course it is, because we used the same prices to compute nominal GDP as in the real GDP computation above.

The growth rate of real GDP from year 1 to year 2 is then equal to

$$g_2^1 = \frac{Y_2^1 - Y_1^1}{Y_1^1} = \frac{1420 - 1400}{1400} \approx 0.014 = 1.4\%$$

where the superscript again denotes the base year. In the same way, we find that $g_3^1 = 2.8\%$.

If we take year 2 as base year, the prices which we use to compute real GDP are 6 for good 1 and 10 for good 2. Real GDP in year 1, 2 and 3 are then equal to

$$\begin{aligned} Y_1^2 &= \sum_i p_{i,2} q_{i,1} = 6 \cdot 100 + 10 \cdot 100 = 1600 \\ Y_2^2 &= \sum_i p_{i,2} q_{i,2} = 6 \cdot 80 + 10 \cdot 110 = 1580 \\ Y_3^2 &= \sum_i p_{i,2} q_{i,3} = 6 \cdot 90 + 10 \cdot 110 = 1640 \end{aligned}$$

and the growth rates of real GDP are $g_2^2 = -1.3\%$ and $g_3^2 = 3.8\%$.

In this way, we can fill in Table 3.

Table 3: Real GDP at constant prices

	Year 1	Year 2	Year 3
Real GDP, prices of year 1			
• level	1400	1420	1460
• growth rate	-	1.4%	2.8%
Real GDP, prices of year 2			
• level	1600	1580	1640
• growth rate	-	-1.3%	3.8%
Real GDP, prices of year 3			
• level	2200	2060	2180
• growth rate	-	-6.4%	5.8%

Note that the *level* of real GDP depends on the choice of the base year: the lower the prices in the base year, the lower the level of real GDP. This is not really a problem, however, because in general, we are not interested in the *level* of real GDP, but in the *growth rate* of real GDP.

Unfortunately, the growth rate of real GDP *also* depends on the base year. Our computations show why this is the case.

Consider, for instance, the growth rate of real GDP between year 1 and year 2. If we take year 1 as base year, good 1 is valued at a very low price (4) compared with good 2 (which is valued at 10). The decrease in production of good 1 between year 1 and year 2 therefore carries a low weight compared with the increase in production of good 2 - which results in a positive growth rate of real GDP. However, if we take year 2 as base year, good 1 is valued at a higher price (6), while the price of good 2 is still the same (10). The decrease in production of good 1 therefore carries a heavier weight in the real GDP computations. It turns out that it even more than offsets the increase in production of good 2 - which results in a *negative* growth rate of real GDP. And real GDP growth is even more negative if we take year 3 as base year: the production of good 1 is then valued at a very high price of 12 (while the production of good 2 is still valued at 10), such that the decrease in production of good 1 between year 1 and year 2 heavily weighs on real GDP growth between year 1 and year 2.

So which base year should we then choose if we want to measure the change in production between year 1 and year 2? There is no good choice. Luckily, economists have invented an alternative real GDP measure: *chain-weighted* real GDP.

2.2 Chain-weighted real GDP

Suppose we want to measure the increase in production between year 1 and year 2. Which year should we choose as base year?

It doesn't make much sense to use the prices of year 3 to compute the increase in production between year 1 and year 2 - especially as in year 3, the relative price of good 1 (compared with the price of good 2) is quite different from the relative price of good 1 in years 1 and 2: in year 3, good 1 is even more expensive than good 2, while it is cheaper than good 2 in years 1 and 2.

But should we choose year 1 or should we choose year 2 as base year? Most of the time, there is no obvious reason for preferring year 1 rather than year 2, or vice versa. So a logical way to proceed is to first compute real GDP growth with year 1 as base year, then with year 2 as base year, and to take the average of both growth rates.

So let us do this. Table 3 shows that real GDP growth between year 1 and year 2 with year 1 as base year is equal to 1.4%, while real GDP growth between year 1 and year 2 with year 2 as base year is equal to -1.3%. Usually, it is a good idea to compute the average of two growth rates with a geometric (rather than an arithmetic) mean, as in the following formula:

$$1 + g_2^{ch} = \sqrt{(1 + g_1^1)(1 + g_2^2)}$$

where the superscript *ch* stands for "chain-weighted" (for reasons which will become clear later on).

From the results in Table 3 follows then that $g_2^{ch} = \sqrt{(1 + 0.014)(1 - 0.013)} - 1 = 0.001 = 0.1\%$. In the same way, we find that $g_3^{ch} = \sqrt{(1 + g_2^2)(1 + g_3^3)} = \sqrt{(1 + 0.038)(1 + 0.058)} - 1 = 0.048 = 4.8\%$.

Or more generally: the chain-weighted growth rate of real GDP between year t and year $t + 1$ is such that

$$1 + g_{t+1}^{ch} = \sqrt{(1 + g_{t+1}^t)(1 + g_{t+1}^{t+1})}$$

Note that if the absolute values of g_{t+1}^t and g_{t+1}^{t+1} are small (compared to 1), g_{t+1}^{ch} is approximately equal to the arithmetic mean:

$$g_{t+1}^{ch} \approx \frac{1}{2}(g_{t+1}^t + g_{t+1}^{t+1})$$

Usually, we are not really interested in the *level* of chain-weighted real GDP, but often it is useful to have a number for it anyway. A logical thing to do then is to nail the level of chain-weighted real GDP to the nominal GDP level in a certain base year.

Let us take year 1 as base year, for instance. From Table 2, we find that nominal GDP in year 1 is equal to 1400. The chain-weighted real GDP level in year 1, with year 1 as base year, is then equal to

$$Y_1^{ch,1} = P_1 Y_1 = 1400$$

where the superscript *ch, 1* refers to the fact $Y_1^{ch,1}$ is a chain-weighted real GDP level with base year 1.

Remember now that the chain-weighted real GDP growth rate from year 1 to year 2 is 0.1%. We can then compute the chain-weighted real GDP level in period 2, with year 1 as base year, as

$$Y_2^{ch,1} = Y_1^{ch,1} \cdot (1 + g_2^{ch}) = 1400 \cdot (1 + 0.001) = 1401.1$$

Once we have $Y_2^{ch,1}$, we can compute in a similar way $Y_3^{ch,1}$:

$$Y_3^{ch,1} = Y_2^{ch,1} \cdot (1 + g_3^{ch}) = 1401.1 \cdot (1 + 0.048) = 1468.5$$

Of course, we can also choose year 2 as base year. We then set the chain-weighted real GDP level in year 2 equal to nominal GDP in year 2 (which we found in Table 2):

$$Y_2^{ch,2} = P_2 Y_2 = 1580$$

From this anchor point, we can compute the chain-weighted real GDP levels in the other years, by using the chain-weighted growth rates which we found earlier:

$$\begin{aligned} Y_1^{ch,2} &= \frac{Y_2^{ch,2}}{1 + g_2^{ch}} = \frac{1580}{1 + 0.001} = 1578.7 \\ Y_3^{ch,2} &= Y_2^{ch,2} \cdot (1 + g_3^{ch}) = 1580 \cdot (1 + 0.048) = 1655.9 \end{aligned}$$

In a similar way, we can compute the chain-weighted real GDP levels if we choose year 3 as base year:

$$\begin{aligned} Y_3^{ch,3} &= P_3 Y_3 = 2180 \\ Y_2^{ch,3} &= \frac{Y_3^{ch,3}}{1 + g_3^{ch}} = \frac{2180}{1 + 0.048} = 2080.0 \\ Y_1^{ch,3} &= \frac{Y_2^{ch,3}}{1 + g_2^{ch}} = \frac{2080.0}{1 + 0.001} = 2078.4 \end{aligned}$$

The chain-weighted growth rate of real GDP and the chain-weighted levels of real GDP for different base years are summarized in Table 4.

Table 4: Chain-weighted real GDP

	Year 1	Year 2	Year 3
Chain-weighted real GDP			
• level, with base year			
* year 1	1400	1401.1	1468.5
* year 2	1578.7	1580	1655.9
* year 3	2078.4	2080.0	2180
• growth rate	-	0.1%	4.8%

So to compute a series of chain-weighted *growth rates* of real GDP, the weights (i.e. the prices) which the quantities of the different goods get in the computations are not fixed (as in the computations of real GDP at constant prices), but change over time. And to construct a series of chain-weighted *levels* of real GDP, we use the series of chain-weighted growth rates to link the level of chain-weighted real GDP in a base period to the levels of chain-weighted real GDP in all other periods. In this way, we make a chain of real GDP levels by using ever changing weights of the produced quantities of the different goods.

3 GDP at Purchasing Power Parity (PPP)

Measuring GDP and keeping track of how a country's production changes over time is difficult. But comparing the production in one country with the production in another country often turns out to be *extremely* difficult. Most international comparisons rely on the methodology which was developed by The World Bank. If you want to know all the ins and outs of this methodology, you'll have to read the 659 pages of *Measuring the Real Size of the World Economy*, a free e-book published by The World Bank. Luckily, the essence of the World Bank's methodology can be illustrated with a simple example (sidestepping all kinds of conceptual intricacies and cutting several mathematically challenging corners).

Suppose we want to compare the production of final goods in The Netherlands (NL) with the production of final goods in the U.S. To make life easier, let us suppose that both The Netherlands and the U.S. produce only two final goods, good 1 and good 2. The quantity and the price of good i in The Netherlands are denoted by $q_{i,NL}$ and $p_{i,NL}$, respectively; and the quantity and the price of good i in the U.S. are denoted by $q_{i,US}$ and $p_{i,US}$. The quantities q and the prices p of the final goods in both countries are given in Table 5.

Table 5: Quantities and prices

	NL	U.S.
Quantity		
• of good 1	15	100
• of good 2	5	300
Price		
• of good 1	€1	\$2
• of good 2	€3	\$1

Note that the prices $p_{1,NL}$ and $p_{2,NL}$ are in euros (€), while the prices $p_{1,US}$ and $p_{2,US}$ are in U.S. dollars (\$). Nominal GDP in The Netherlands and the U.S. is then equal to

$$P_{NL}Y_{NL} = \sum_i p_{i,NL}q_{i,NL} = p_{1,NL}q_{1,NL} + p_{2,NL}q_{2,NL} = €1 \cdot 15 + €3 \cdot 5 = €30$$

$$P_{US}Y_{US} = \sum_i p_{i,US}q_{i,US} = p_{1,US}q_{1,US} + p_{2,US}q_{2,US} = \$2 \cdot 100 + \$1 \cdot 300 = \$500$$

In order to be able to compare GDP in The Netherlands with GDP in the U.S., we need to express both GDP levels in a common currency.

One option is to use the market exchange rate between the euro and the U.S. dollar to convert GDP in The Netherlands into U.S. dollars (or vice versa, to convert GDP in the U.S. into euros). But this is a bad idea. First of all, market exchange rates are quite volatile. It doesn't make much sense to compare the production levels in different countries in a way that substantially changes from one day to another because of day-to-day exchange rate fluctuations. The second problem with market exchange rates is that even though they may be useful to compare the value of goods and services that are traded internationally, they are less useful to compare the value of goods and services that are consumed in the same country as where they are produced (such as haircuts). It turns out that in poor countries, goods and services that are *not* traded internationally are relatively cheap, while in rich countries the same goods and services are often much more expensive. That is why Mexicans who travel to the U.S. prefer to have their hair cut in Mexico before they leave rather than in the U.S. when they arrive, and why Dutch people - who are on a diet of expensive "broodje kroket" sandwiches in The Netherlands - indulge in sumptuous meals when they are holidaying in sunnier but less prosperous countries. Using market exchange rates to compare the value of the production in poor countries with the value of the production in richer countries would then underestimate the purchasing power of GDP in poor countries compared with richer countries.

Economists therefore take a different approach, which focuses on the purchasing power of GDP. In a nutshell, it goes as follows.

First take a "benchmark" country, for which economists usually choose the U.S. Now split the complete basket of goods and services that has been produced in the U.S. in a given year in many identical baskets, each worth exactly \$1. As in our example GDP in the U.S. is \$500, there would be 500 of such baskets, each consisting of $q_{1,US}/500 = 100/500 = 1/5$ units of good 1 and $q_{2,US}/500 = 300/500 = 3/5$ units of good 2.

Now introduce a fictitious currency (i.e. a currency that does not exist in the real world), called the *international dollar*, with symbol I\$. Assume that one international dollar buys exactly one of these tiny baskets of $1/5$ units of good 1 and $3/5$ units of good 2. As the total production in the U.S. consists of 500 of such baskets, GDP in the U.S. expressed in I\$ is then simply equal to I\$ 500:

$$P_{US}Y_{US} = \text{I\$ } 500$$

We now convert GDP in The Netherlands into I\$. In The Netherlands, the price of one tiny basket of $1/5$ units of good 1 and $3/5$ units of good 2 is $p_{1,NL} \cdot 1/5 + p_{2,NL} \cdot 3/5 = \text{€}1 \cdot 1/5 + \text{€}3 \cdot 3/5 = \text{€}2$. So two euro is worth the same as one international dollar. Dividing Dutch GDP (expressed in euro) by €2 per I\$ yields then GDP in The Netherlands expressed in I\$:

$$P_{NL}Y_{NL} = \frac{\text{€}30}{\text{€}2/\text{I\$}} = \text{I\$ } 15$$

We can then compare GDP in The Netherlands in I\$ with GDP in the U.S. in I\$. In this way, we compare GDP in The Netherlands with GDP in the U.S. by expressing both GDP levels in a common currency which has the same purchasing power in both countries. GDP in I\$ is therefore also called "GDP at Purchasing Power Parity (PPP)".

To extend this analysis over time, economists usually take the prices of the benchmark country in a base year, and then compute GDP of different countries in different years, all expressed in I\$ in prices of this base year.